



## Microcontrollers

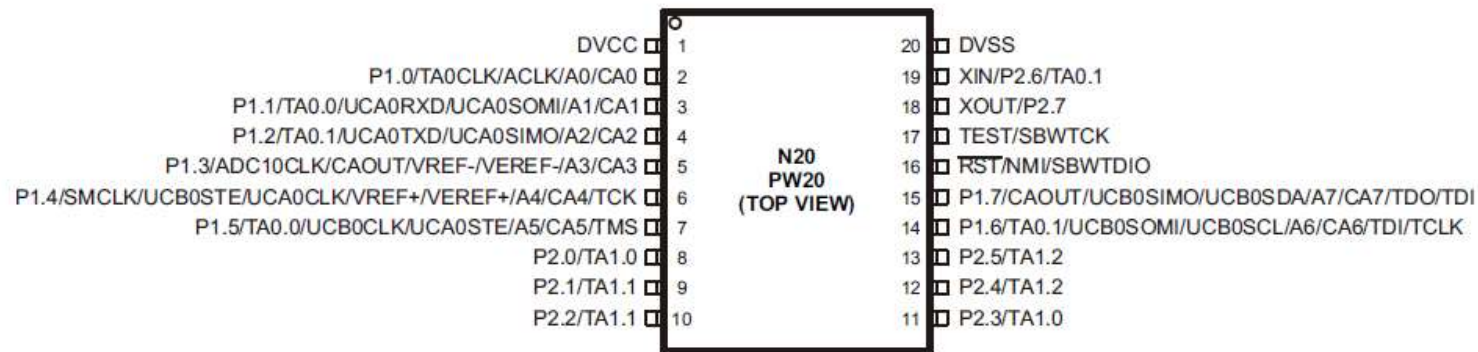
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**Rex Joseph**

Department of Electrical and  
Electronics Engineering

## Memory organization Flash

Device Pinout, MSP430G2x13 and MSP430G2x53, 20-Pin Devices, TSSOP and PDIP

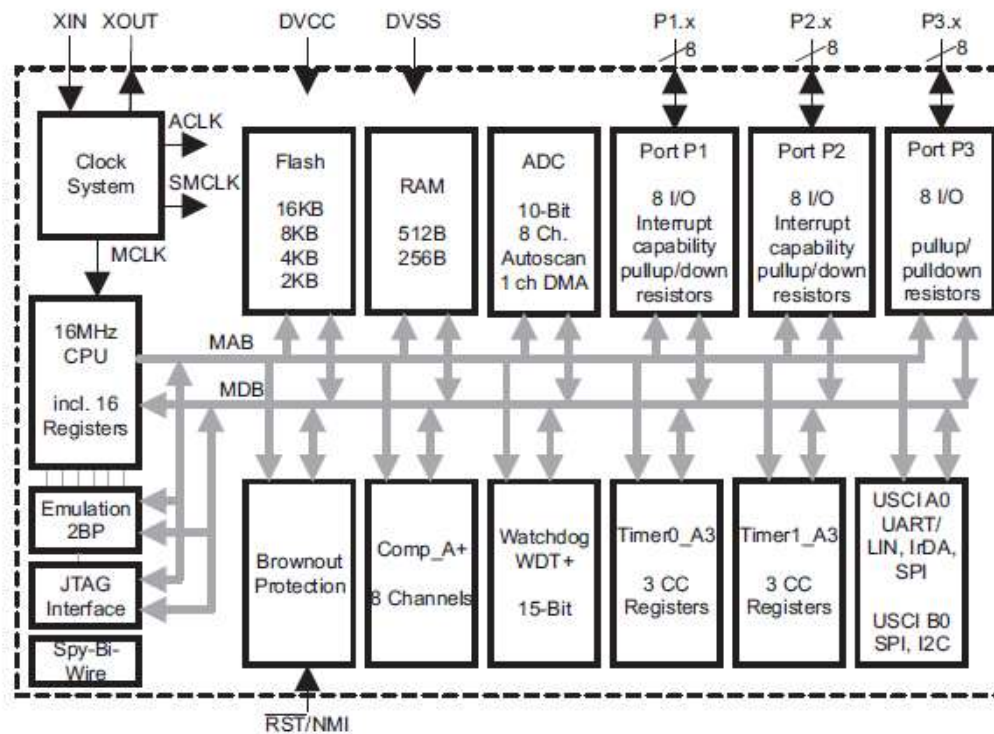


NOTE: ADC10 is available on MSP430G2x53 devices only.

NOTE: The pulldown resistors of port P3 should be enabled by setting P3REN.x = 1.

## Memory organization Flash

Functional Block Diagram, MSP430G2x53



NOTE: Port P3 is available on 28-pin and 32-pin devices only.

## Memory organization/programming Flash

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The MSP430 flash memory is bit-, byte-, and word-addressable and programmable.

The flash memory module has an integrated controller that controls programming and erase operations.

The controller has four registers, a timing generator, and a voltage generator to supply program and erase voltages.

MSP430 flash memory features include:

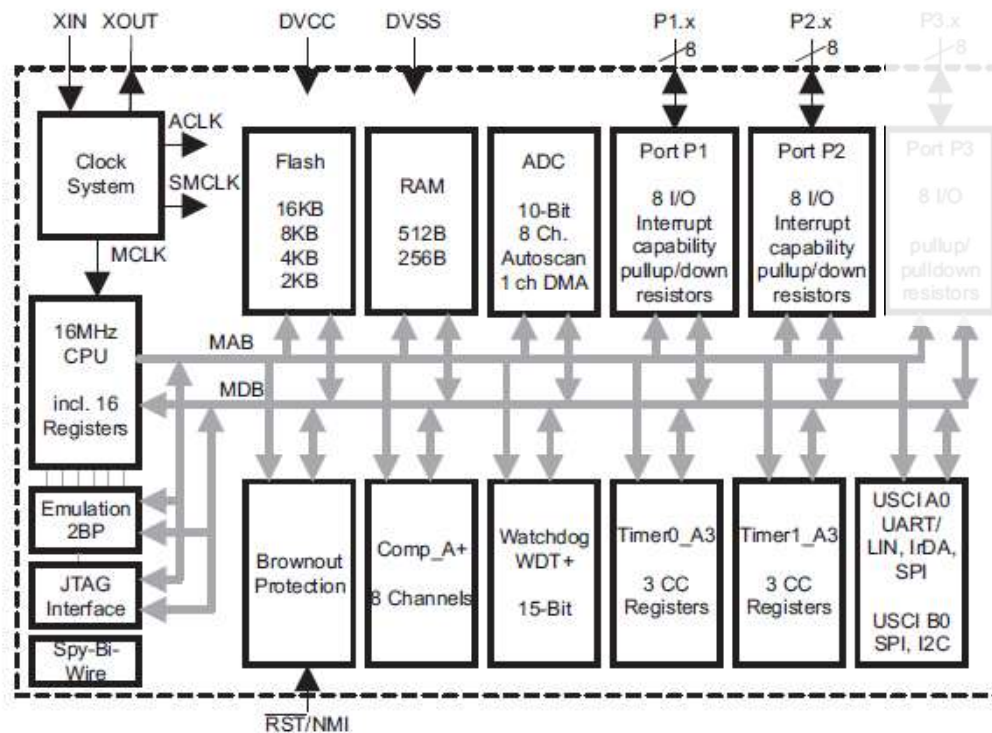
- Internal **programming voltage** generation
- Bit, byte, or word programmable
- Ultra-low-power operation
- Segment erase and mass erase
- Marginal 0 and marginal 1 read mode (**optional, device-specific**)

# Memory organization Flash

Functional Block Diagram, MSP430G2x53

16-Bit Hex - TI Style

```
0x012A 9642
0x012C Flash_FCTL3
0x012C 9658
0x012E Timer0_A3_TA0IV
0x012E 0000 0000 0000 0000 0000
0x0142 0000 0000 0000 0000 0000
0x0160 Timer0_A3_TA0CTL
0x0160 0000
0x0162 Timer0_A3_TA0CCTL0
0x0162 0000
0x0164 Timer0_A3_TA0CCTL1
0x0164 0000
0x0166 Timer0_A3_TA0CCTL2
0x0166 0400 0000 0000 0000 0000
0x0170 Timer0_A3_TA0R
0x0170 0000
0x0172 Timer0_A3_TA0CCR0
0x0172 0000
0x0174 Timer0_A3_TA0CCR1
0x0174 0000
0x0176 Timer0_A3_TA0CCR2
0x0176 0000 0000 0000 0000 0000
0x0180 Timer1_A3_TA1CTL
0x0180 0000
```



16-Bit Hex - TI Style

```
0x0182 Timer1_A3_TA1CCTL0
0x0182 0400
0x0184 Timer1_A3_TA1CCTL1
0x0184 0000
0x0186 Timer1_A3_TA1CCTL2
0x0186 0400 0000 0000 0000 0000
0x0190 Timer1_A3_TA1R
0x0190 0000
0x0192 Timer1_A3_TA1CCR0
0x0192 0000
0x0194 Timer1_A3_TA1CCR1
0x0194 0000
0x0196 Timer1_A3_TA1CCR2
0x0196 0000 0000 0000 0000 0000
0x01B0 ADC10_ADC10CTL0
0x01B0 0000
0x01B2 ADC10_ADC10CTL1
0x01B2 0000
0x01B4 ADC10_ADC10MEM
0x01B4 0000 0000 0000 0000
0x01BC ADC10_ADC10SA
0x01BC 0200 0000 0000 0000 0000
0x01DA 0000 0000 0000 0000 0000
```

NOTE: Port P3 is available on 28-pin and 32-pin devices only.



# Memory organization Flash

| Device            | BSL | EEM | Flash (KB) | RAM (B) | Timer_A | COMP_A+ Channel | ADC10 Channel | USCI_A0, USCI_B0 | Clock        | I/O | Package Type |
|-------------------|-----|-----|------------|---------|---------|-----------------|---------------|------------------|--------------|-----|--------------|
| MSP430G2553IRHB32 | 1   | 1   | 16         | 512     | 2x TA3  | 8               | 8             | 1                | LF, DCO, VLO | 24  | 32-QFN       |
| MSP430G2553IPW28  |     |     |            |         |         |                 |               |                  |              | 24  | 28-TSSOP     |
| MSP430G2553IPW20  |     |     |            |         |         |                 |               |                  |              | 16  | 20-TSSOP     |
| MSP430G2553IN20   |     |     |            |         |         |                 |               |                  |              | 16  | 20-PDIP      |
| MSP430G2453IRHB32 | 1   | 1   | 8          | 512     | 2x TA3  | 8               | 8             | 1                | LF, DCO, VLO | 24  | 32-QFN       |
| MSP430G2453IPW28  |     |     |            |         |         |                 |               |                  |              | 24  | 28-TSSOP     |
| MSP430G2453IPW20  |     |     |            |         |         |                 |               |                  |              | 16  | 20-TSSOP     |
| MSP430G2453IN20   |     |     |            |         |         |                 |               |                  |              | 16  | 20-PDIP      |



MSP430G2x53  
MSP430G2x13

Does it  
add up?

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SLAS735J – APRIL 2011 – REVISED MAY 2013

## MIXED SIGNAL MICROCONTROLLER

```

0x10A2  FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF
0x10C0  TLV_Calibration_Data_TLV_CHECKSUM
0x10C0  8E93 16FE FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF
0x10DA  TLV_Calibration_Data_TLV_ADC10_1_TAG
0x10DA  1010 8000 0000 7D8A 02E4 0371 7DF7 01BA 020D 08FE FFFF FFFF FFFF
0x10F6  TLV_Calibration_Data_TLV_DCO_30_TAG
0x10F6  0801
0x10F8  Calibration_Data_CALDCO_16MHZ
0x10F8  8F81
0x10FA  Calibration_Data_CALDCO_12MHZ
0x10FA  8E87
0x10FC  Calibration_Data_CALDCO_8MHZ
0x10FC  8D6D
0x10FE  Calibration_Data_CALDCO_1MHZ
0x10FE  86B7 3FFF 3FFF 3FFF 3FFF 3FFF 3FFF

```

```

0x8FE2  3FFF 3FFF 3FFF 3FFF 3FFF 3FFF 3FFF 3FFF
0xC000  $./main.asm:19:65$, RESET
0xC000  4031 0400
0xC004  StopWDT
0xC004  40B2 5A80 0120 43D2 0022
0xC00E  Mainloop
0xC00E  403F 02F0
0xC012  L1
0xC012  43D2
0xC014  0021 4F0E 12B0 C054 43C2 0021 403E 1E00
0xC032  L3
0xC032  43D2 0021
0xC036  4F0E 12B0 C054 43C2 0021 403E 1E00 8F00
0xC054  L2
0xC054  422D 8D0E 37FD 4130
0xC05C  $isr_trap.asm:48:59$, __TI_ISR_TRAP
0xC05C  D032 0010 3FFD 4303 FFFF FFFF FFFF

```

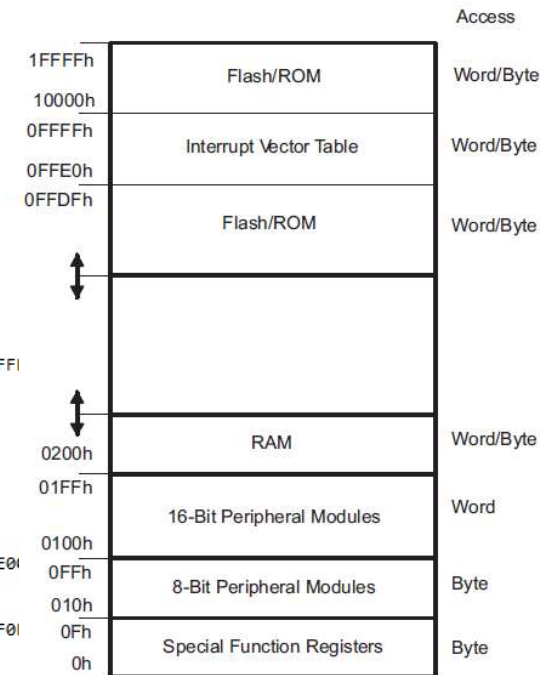


Figure 1-2. Memory Map

# MSP430G2553 memory

## Memory Organization

**Table 8. Memory Organization**

|                        |           | MSP430G2153      | MSP430G2253<br>MSP430G2213 | MSP430G2353<br>MSP430G2313 | MSP430G2453<br>MSP430G2413 | MSP430G2553<br>MSP430G2513 |
|------------------------|-----------|------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| Memory                 | Size      | 1kB              | 2kB                        | 4kB                        | 8kB                        | 16kB                       |
| Main: interrupt vector | Flash     | 0xFFFF to 0xFFC0 | 0xFFFF to 0xFFC0           | 0xFFFF to 0xFFC0           | 0xFFFF to 0xFFC0           | 0xFFFF to 0xFFC0           |
| Main: code memory      | Flash     |                  |                            |                            |                            |                            |
| Information memory     | Size      | 256 Byte         | 256 Byte                   | 256 Byte                   | 256 Byte                   | 256 Byte                   |
|                        | Flash     | 010FFh to 01000h | 010FFh to 01000h           | 010FFh to 01000h           | 010FFh to 01000h           | 010FFh to 01000h           |
| RAM                    | Size      | 256 Byte         | 256 Byte                   | 256 Byte                   | 512 Byte                   | 512 Byte                   |
| Peripherals            | 16-bit    | 01FFh to 0100h   | 01FFh to 0100h             | 01FFh to 0100h             | 01FFh to 0100h             | 01FFh to 0100h             |
|                        | 8-bit     | 0FFh to 010h     | 0FFh to 010h               | 0FFh to 010h               | 0FFh to 010h               | 0FFh to 010h               |
|                        | 8-bit SFR | 0Fh to 00h       | 0Fh to 00h                 | 0Fh to 00h                 | 0Fh to 00h                 | 0Fh to 00h                 |

# Memory organization Flash

**Table 8. Memory Organization**

|                        |           | MSP430G2153      | MSP430G2253<br>MSP430G2213 | MSP430G2353<br>MSP430G2313 | MSP430G2453<br>MSP430G2413 | MSP430G2553<br>MSP430G2513 |
|------------------------|-----------|------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| Memory                 | Size      | 1kB              | 2kB                        | 4kB                        | 8kB                        | 16kB                       |
| Main: interrupt vector | Flash     | 0xFFFF to 0xFFC0 | 0xFFFF to 0xFFC0           | 0xFFFF to 0xFFC0           | 0xFFFF to 0xFFC0           | 0xFFFF to 0xFFC0           |
| Main: code memory      | Flash     | 0xFFFF to 0xFC00 | 0xFFFF to 0xF800           | 0xFFFF to 0xF000           | 0xFFFF to 0xE000           | 0xFFFF to 0xC000           |
| Information memory     | Size      | 256 Byte         | 256 Byte                   | 256 Byte                   | 256 Byte                   | 256 Byte                   |
|                        | Flash     | 010FFh to 01000h | 010FFh to 01000h           | 010FFh to 01000h           | 010FFh to 01000h           | 010FFh to 01000h           |
| RAM                    | Size      | 256 Byte         | 256 Byte                   | 256 Byte                   | 512 Byte                   | 512 Byte                   |
|                        |           | 0x02FF to 0x0200 | 0x02FF to 0x0200           | 0x02FF to 0x0200           | 0x03FF to 0x0200           | 0x03FF to 0x0200           |
| Peripherals            | 16-bit    | 01FFh to 0100h   | 01FFh to 0100h             | 01FFh to 0100h             | 01FFh to 0100h             | 01FFh to 0100h             |
|                        | 8-bit     | 0FFh to 010h     | 0FFh to 010h               | 0FFh to 010h               | 0FFh to 010h               | 0FFh to 010h               |
|                        | 8-bit SFR | 0Fh to 00h       | 0Fh to 00h                 | 0Fh to 00h                 | 0Fh to 00h                 | 0Fh to 00h                 |



## Memory organization/programming Flash

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### Flash Memory

The flash memory can be programmed via the Spy-Bi-Wire/JTAG port or in-system by the CPU.

- Flash memory has n segments of main memory and **four segments of information memory (A to D) of 64 bytes each**. Each segment in main memory is 512 bytes in size.
- Segments 0 to n may be erased in one step, or each segment may be individually erased.
- Segments A to D can be erased individually or as a group with segments 0 to n. Segments A to D are also called *information memory*.
- Segment A contains **calibration data**. After reset **segment A** is **protected** against programming and erasing.

It can be unlocked but care should be taken not to erase this segment if the device-specific calibration data is required.

## Memory organization/programming Flash

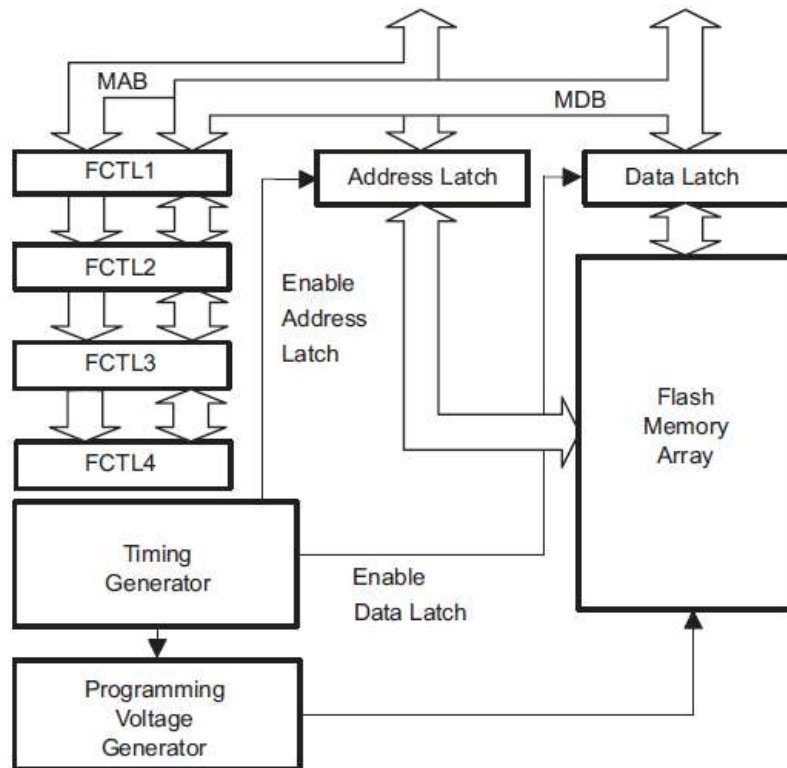


Figure 7-1. Flash Memory Module Block Diagram

### Minimum VCC during flash write or erase

The minimum VCC voltage during a flash write or erase operation is 2.2 V.

If VCC falls below 2.2 V during write or erase, the result of the write or erase is unpredictable.

## Memory organization/programming Flash

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### Flash Memory Segmentation

MSP430 flash memory is partitioned into segments.

Single bits, bytes, or words can be written to flash memory, but the segment is the smallest size of flash memory that can be **erased**.

The flash memory is partitioned into **main** and **information memory** sections.

There is no difference in the operation of the main and information memory sections. Code or data can be located in either section.

The differences between the two sections are the segment size and the physical addresses.

## Memory organization/programming Flash

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The information memory has four 64-byte **segments**.

The main memory has one or more 512-byte segments.

The main memory segments are further divided into **blocks** of 64 bytes.  
(device-specific)

# Memory organization/programming Flash

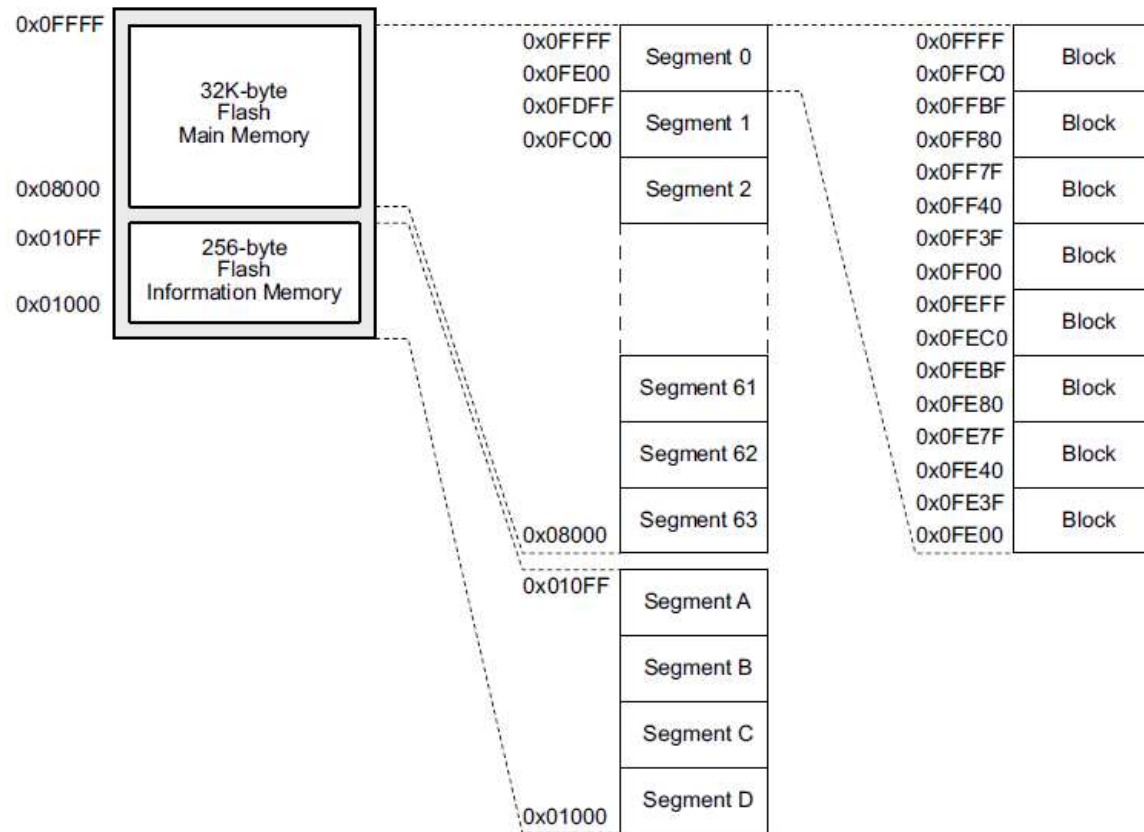


Figure 7-2. Flash Memory Segments, 32KB Example



## Memory organization/programming Flash

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### Segment A

Segment A of the information memory is locked separately from all other segments with the LOCKA bit.

When LOCKA = 1, Segment A cannot be written or erased and all information memory is protected from erasure during a mass erase or production programming.

When LOCKA = 0, Segment A can be erased and written as any other flash memory segment, and all information memory is erased during a mass erase or production programming.

The state of the LOCKA bit is toggled when a 1 is written to it. Writing a 0 to LOCKA has no effect.

This allows existing flash programming routines to be used unchanged.

## Memory organization/programming Flash

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|               |                                                                  |                                                                                                                                                   |
|---------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| SEGA_UNLOCKED | BIT #LOCKA,&FCTL3<br>JZ SEGA_UNLOCKED<br>MOV #FWKEY+LOCKA,&FCTL3 | ; Unlock SegmentA<br>; Test LOCKA<br>; Already unlocked?<br>; No, unlock SegmentA<br>; Yes, continue<br>; SegmentA is unlocked<br>; Lock SegmentA |
| SEGA_LOCKED   | BIT #LOCKA,&FCTL3<br>JNZ SEGA_LOCKED<br>MOV #FWKEY+LOCKA,&FCTL3  | ; Test LOCKA<br>; Already locked?<br>; No, lock SegmentA<br>; Yes, continue<br>; SegmentA is locked                                               |

## Memory organization/programming Flash

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### Flash Memory Operation

The **default** mode of the flash memory is **read mode**.

In read mode, the flash memory is not being erased or written, the flash timing generator and voltage generator are off, and the memory operates identically to ROM.

MSP430 flash memory is in-system programmable (ISP) without the need for additional external voltage.

The CPU can program its own flash memory.

## Memory organization/programming Flash

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The flash memory write and erase modes are selected with the BLKWRT, WRT, MERAS, and ERASE bits and are:

- Byte or word write
- Block write
- Segment erase
- Mass erase (all main memory segments)
- All erase (all segments)

Reading from or writing to flash memory while it is being programmed or erased is prohibited.

If CPU execution is required during the write or erase, the code to be executed must be in RAM.

Any flash update can be initiated from within flash memory or RAM.

## Memory organization/programming Flash

### Erasing Flash Memory

The erased level of a flash memory bit is 1.

Each bit can be programmed from **1 to 0 individually** but to reprogram from **0 to 1** requires an **erase cycle**.

The smallest amount of flash that can be erased is a **segment**.

There are three erase modes selected with the ERASE and MERAS bits

**Table 7-1. Erase Modes**

| MERAS | ERASE | Erase Mode                                                                                      |
|-------|-------|-------------------------------------------------------------------------------------------------|
| 0     | 1     | Segment erase                                                                                   |
| 1     | 0     | Mass erase (all main memory segments)                                                           |
| 1     | 1     | LOCKA = 0: Erase main and information flash memory.<br>LOCKA = 1: Erase only main flash memory. |

## Memory organization/programming Flash

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Any erase is **initiated** by a dummy write into the address range to be erased.

The dummy write starts the flash **timing generator** and the **erase operation**.

The BUSY bit is set immediately after the dummy write and remains set throughout the erase cycle.

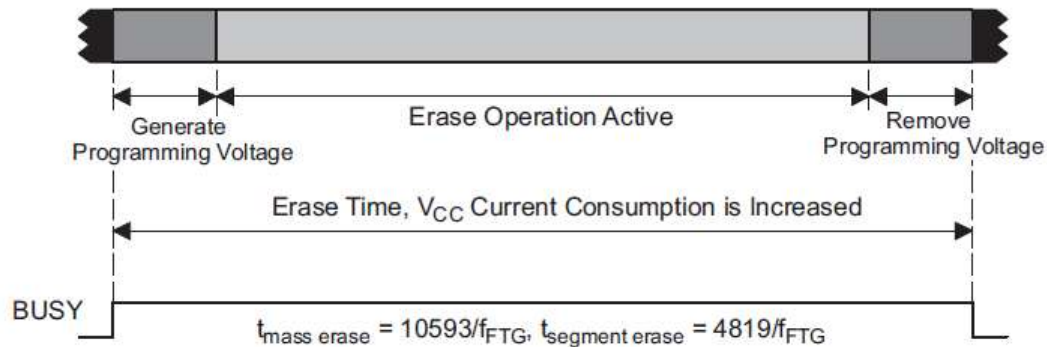
BUSY, MERAS, and ERASE are automatically cleared when the cycle completes.

The erase cycle timing is **not dependent on the amount of flash memory** present on a device.

Erase cycle times are **equivalent for all MSP430F2xx and MSP430G2xx devices**.



## Memory organization/programming Flash



**Figure 7-4. Erase Cycle Timing**

A dummy write to an address not in the range to be erased does not start the erase cycle, does not affect the flash memory, and is not flagged in any way.

This errant dummy write is ignored.

## Memory organization/programming Flash

### *Initiating an Erase From Within Flash Memory*

Any erase cycle can be initiated from **within flash** memory or from **RAM**.

When a flash segment erase operation is initiated from within flash memory, all timing is controlled by the flash controller, and the **CPU is held** while the erase cycle completes.

After the erase cycle completes, the CPU resumes code execution with the instruction following the dummy write.

When initiating an erase cycle from within flash memory, it is possible to erase the code needed for execution after the erase.

If this occurs, CPU execution is unpredictable after the erase cycle.

## Memory organization/programming Flash

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### *Initiating an Erase From RAM*

Any erase cycle may be initiated from RAM.

In this case, the CPU is not held and can continue to execute code from RAM.

The BUSY bit must be polled to determine the end of the erase cycle before the CPU can access any flash address again.

If a flash access occurs while  $BUSY = 1$ , it is an access violation, ACCVIFG is set, and the erase results are unpredictable.

## Memory organization/programming Flash

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### ***Block Write***

The block write can be used to accelerate the flash write process when many sequential bytes or words need to be programmed.

The flash programming voltage remains on for the duration of writing the 64-byte block.

The cumulative programming time  $t_{CPT}$  must not be exceeded for any block during a block write.

A block write **cannot be initiated from within flash memory.**

The block write must be initiated from RAM only.

The BUSY bit remains set throughout the duration of the block write.

## Memory organization/programming Flash

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### ***Block Write***

The WAIT bit must be checked between writing each byte or word in the block.

When WAIT is reset the next byte or word of the block can be written.

When writing successive blocks, the BLKWRT bit must be cleared after the current block is complete.

BLKWRT can be set initiating the next block write after the required flash recovery time.

BUSY is cleared following each block write completion indicating the next block can be written

## Memory organization/programming Flash

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### Flash Memory Access During Write or Erase

When any write or any erase operation is initiated from **RAM** and while  $BUSY = 1$ , the CPU may not read or write to or from any flash location.

Otherwise, an access violation occurs,  $ACCVIFG$  is set, and the result is unpredictable.

Also if a write to flash is attempted with  $WRT = 0$ , the  $ACCVIFG$  interrupt flag is set, and the flash memory is unaffected.

## Memory organization/programming Flash

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When a byte or word write or any erase operation is initiated from within flash memory, the flash controller returns op-code 03FFFh to the CPU at the next instruction fetch.

Op-code 03FFFh is the JMP PC instruction.

This causes the CPU to loop until the flash operation is finished.

When the operation is finished and  $BUSY = 0$ , the flash controller allows the CPU to fetch the proper op-code and program execution resumes.

## Memory organization/programming Flash

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### Programming Flash Memory Devices

There are three options for programming an MSP430 flash device. All options support in-system programming:

- Program with JTAG
- Program with the bootloader
- Program with a custom solution



## Memory organization/programming Flash

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### ***Programming Flash Memory With JTAG***

MSP430 devices can be programmed through the JTAG port. The JTAG interface requires four signals (five signals on 20- and 28-pin devices), ground and, optionally, VCC and RST/NMI.

The JTAG port is protected with a fuse.

Blowing the fuse completely disables the JTAG port and is not reversible.

Further access to the device through JTAG is not possible.

## Memory organization/programming Flash

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### ***Programming Flash Memory With the Bootloader (BSL)***

Most MSP430 flash devices contain a bootloader.

The BSL enables users to read or program the flash memory or RAM using a UART serial interface.

Access to the MSP430 flash memory through the BSL is protected by a 256-bit user-defined password.

## Memory organization/programming Flash

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### ***Programming Flash Memory With a Custom Solution***

The ability of the MSP430 CPU to write to its own flash memory allows for in-system and external custom programming solutions

The user can choose to provide data to the MSP430 through any means available (for example, UART or SPI).

User-developed software can receive the data and program the flash memory.

Because this type of solution is developed by the user, it can be completely customized to fit the application needs for programming, erasing, or updating the flash memory

## Memory organization/programming Flash

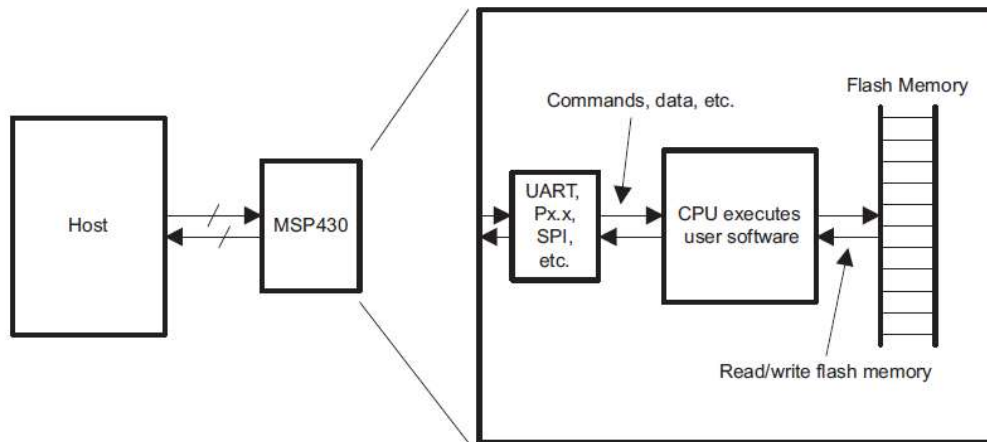


Figure 7-12. User-Developed Programming Solution

Table 7-4. Flash Memory Registers

| Address | Acronym              | Register Name          | Type       | Reset                         | Section                       |
|---------|----------------------|------------------------|------------|-------------------------------|-------------------------------|
| 128h    | FCTL1                | Flash memory control 1 | Read/write | 9600h with PUC                | <a href="#">Section 7.4.1</a> |
| 12Ah    | FCTL2                | Flash memory control 2 | Read/write | 9642h with PUC                | <a href="#">Section 7.4.2</a> |
| 12Ch    | FCTL3                | Flash memory control 3 | Read/write | 9658h with PUC <sup>(1)</sup> | <a href="#">Section 7.4.3</a> |
| 1BEh    | FCTL4 <sup>(2)</sup> | Flash memory control 4 | Read/write | 00h with PUC                  | <a href="#">Section 7.4.4</a> |
| 0h      | IE1                  | Interrupt enable 1     | Read/write | 00h with PUC                  | <a href="#">Section 7.4.5</a> |

(1) KEYV is initialized with POR. All other bits are initialized with PUC.

(2) Not present in all devices. See device-specific data sheet.

## Flash Control Registers

**Figure 7-13. FCTL1 Register**

| 15             | 14   | 13       | 12                   | 11                 | 10    | 9     | 8        |
|----------------|------|----------|----------------------|--------------------|-------|-------|----------|
| FRKEY<br>FWKEY |      |          |                      |                    |       |       |          |
| rw-1           | rw-0 | rw-0     | rw-1                 | rw-0               | rw-1  | rw-1  | rw-0     |
| 7              | 6    | 5        | 4                    | 3                  | 2     | 1     | 0        |
| BLKWRT         | WRT  | Reserved | EEIEX <sup>(1)</sup> | EEI <sup>(1)</sup> | MERAS | ERASE | Reserved |
| rw-0           | rw-0 | r0       | rw-0                 | rw-0               | rw-0  | rw-0  | r0       |

(1) Not present on MSP430x20xx and MSP430G2xx devices.

## Flash Control Registers

**Table 7-5. FCTL1 Register Field Descriptions**

| Bit  | Field          | Type | Reset | Description                                                                                                                                                                                                                                                                                                  |
|------|----------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | FRKEY<br>FWKEY | R/W  | 96h   | FCTLx password. Always reads as 096h. Must be written as 0A5h. Writing any other value generates a PUC.                                                                                                                                                                                                      |
| 7    | BLKWRT         | R/W  | 0h    | Block write mode. WRT must also be set for block write mode. BLKWRT is automatically reset when EMEX is set.<br>0b = Block-write mode is off<br>1b = Block-write mode is on                                                                                                                                  |
| 6    | WRT            | R/W  | 0h    | Write. This bit is used to select any write mode. WRT is automatically reset when EMEX is set.<br>0b = Write mode is off<br>1b = Write mode is on                                                                                                                                                            |
| 5    | Reserved       | R    | 0h    |                                                                                                                                                                                                                                                                                                              |
| 4    | EEIEX          | R/W  | 0h    | Enable emergency interrupt exit. Setting this bit enables an interrupt to cause an emergency exit from a flash operation when GIE = 1. EEIEX is automatically reset when EMEX is set. Not present on MSP430x20xx and MSP430G2xx devices.<br>0b = Exit interrupt disabled.<br>1b = Exit on interrupt enabled. |

## Flash Control Registers

|   |          |     |    |                                                                                                                                                                                                                                                                                                                                                     |
|---|----------|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | EEIEX    | R/W | 0h | <p>Enable emergency interrupt exit. Setting this bit enables an interrupt to cause an emergency exit from a flash operation when GIE = 1. EEIEX is automatically reset when EMEX is set.</p> <p>Not present on MSP430x20xx and MSP430G2xx devices.</p> <p>0b = Exit interrupt disabled.</p> <p>1b = Exit on interrupt enabled.</p>                  |
| 3 | EEI      | R/W | 0h | <p>Enable erase interrupts. Setting this bit allows a segment erase to be interrupted by an interrupt request. After the interrupt is serviced the erase cycle is resumed.</p> <p>Not present on MSP430x20xx and MSP430G2xx devices.</p> <p>0b = Interrupts during segment erase disabled.</p> <p>1b = Interrupts during segment erase enabled.</p> |
| 2 | MERAS    | R/W | 0h | <p>Mass erase and erase. These bits are used together to select the erase mode. MERAS and ERASE are automatically reset when EMEX is set. See <a href="#">Table 7-6</a>.</p>                                                                                                                                                                        |
| 1 | ERASE    | R/W | 0h |                                                                                                                                                                                                                                                                                                                                                     |
| 0 | Reserved | R   | 0h |                                                                                                                                                                                                                                                                                                                                                     |

## Flash Control Registers

**Figure 7-14. FCTL2 Register**

|        |      |      |      |      |      |      |      |
|--------|------|------|------|------|------|------|------|
| 15     | 14   | 13   | 12   | 11   | 10   | 9    | 8    |
| FWKEYx |      |      |      |      |      |      |      |
| rw-1   | rw-0 | rw-0 | rw-1 | rw-0 | rw-1 | rw-1 | rw-0 |
| 7      | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
| FSSELx |      | FNx  |      |      |      |      |      |
| rw-0   | rw-1 | rw-0 | rw-0 | rw-0 | rw-0 | rw-1 | rw-0 |

**Table 7-7. FCTL2 Register Field Descriptions**

| Bit  | Field  | Type | Reset | Description                                                                                                                                                                                                        |
|------|--------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | FWKEYx | R/W  | 96h   | FCTLx password. Always reads as 096h. Must be written as 0A5h. Writing any other value generates a PUC.                                                                                                            |
| 7-6  | FSSELx | R/W  | 1h    | Flash controller clock source select<br>00b = ACLK<br>01b = MCLK<br>10b = SMCLK<br>11b = SMCLK                                                                                                                     |
| 5-0  | FNx    | R/W  | 2h    | Flash controller clock divider. These six bits select the divider for the flash controller clock. The divisor value is FNx + 1. For example, when FNx = 00h, the divisor is 1. When FNx = 03Fh, the divisor is 64. |



## Flash Control Registers

**Figure 7-15. FCTL3 Register**

|        |        |      |      |      |         |        |      |
|--------|--------|------|------|------|---------|--------|------|
| 15     | 14     | 13   | 12   | 11   | 10      | 9      | 8    |
| FWKEYx |        |      |      |      |         |        |      |
| rw-1   | rw-0   | rw-0 | rw-1 | rw-0 | rw-1    | rw-1   | rw-0 |
| 7      | 6      | 5    | 4    | 3    | 2       | 1      | 0    |
| FAIL   | LOCKA  | EMEX | LOCK | WAIT | ACCVIFG | KEYV   | BUSY |
| r(w)-0 | r(w)-1 | rw-0 | rw-1 | r-1  | rw-0    | rw-(0) | r-0  |

**Table 7-8. FCTL3 Register Field Descriptions**

| Bit  | Field  | Type | Reset | Description                                                                                                                                                                                                                                                                          |
|------|--------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | FWKEYx | R/W  | 96h   | FCTLx password. Always reads as 096h. Must be written as 0A5h. Writing any other value generates a PUC.                                                                                                                                                                              |
| 7    | FAIL   | R/W  | 0h    | Operation failure. This bit is set if the $f_{FTG}$ clock source fails, or a flash operation is aborted from an interrupt when $EEIEX = 1$ . FAIL must be reset with software.<br>0b = No failure<br>1b = Failure                                                                    |
| 6    | LOCKA  | R/W  | 1h    | Segment A and Info lock. Write a 1 to this bit to change its state. Writing 0 has no effect.<br>0b = Segment A unlocked and all information memory is erased during a mass erase.<br>1b = Segment A locked and all information memory is protected from erasure during a mass erase. |

## Flash Control Registers

|   |         |     |    |                                                                                                                                                                                                                                                                                                                                                           |
|---|---------|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | EMEX    | R/W | 0h | Emergency exit<br>0b = No emergency exit<br>1b = Emergency exit                                                                                                                                                                                                                                                                                           |
| 4 | LOCK    | R/W | 1h | Lock. This bit unlocks the flash memory for writing or erasing. The LOCK bit can be set any time during a byte or word write or erase operation, and the operation completes normally. In the block write mode if the LOCK bit is set while BLKWRT = WAIT = 1, then BLKWRT and WAIT are reset and the mode ends normally.<br>0b = Unlocked<br>1b = Locked |
| 3 | WAIT    | R   | 1h | Wait. Indicates the flash memory is being written to.<br>0b = The flash memory is not ready for the next byte or word write<br>1b = The flash memory is ready for the next byte or word write                                                                                                                                                             |
| 2 | ACCVIFG | R/W | 0h | Access violation interrupt flag<br>0b = No interrupt pending<br>1b = Interrupt pending                                                                                                                                                                                                                                                                    |
| 1 | KEYV    | R/W | 0h | Flash security key violation. This bit indicates an incorrect FCTLx password was written to any flash control register and generates a PUC when set. KEYV must be reset with software. KEYV is reset with POR.<br>0b = FCTLx password was written correctly<br>1b = FCTLx password was written incorrectly                                                |

## Flash Control Registers

| 5   | EMEX    | R/W  | 0h    | Emergency exit<br>0b = No emergency exit<br>1b = Emergency exit                                                                                                                                                                                                                                                                                           |
|-----|---------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | LOCK    | R/W  | 1h    | Lock. This bit unlocks the flash memory for writing or erasing. The LOCK bit can be set any time during a byte or word write or erase operation, and the operation completes normally. In the block write mode if the LOCK bit is set while BLKWRT = WAIT = 1, then BLKWRT and WAIT are reset and the mode ends normally.<br>0b = Unlocked<br>1b = Locked |
| 3   | WAIT    | R    | 1h    | Wait. Indicates the flash memory is being written to.<br>0b = The flash memory is not ready for the next byte or word write<br>1b = The flash memory is ready for the next byte or word write                                                                                                                                                             |
| 2   | ACCVIFG | R/W  | 0h    | Access violation interrupt flag<br>0b = No interrupt pending<br>1b = Interrupt pending                                                                                                                                                                                                                                                                    |
| 1   | KEYV    | R/W  | 0h    | Flash security key violation. This bit indicates an incorrect FCTLx password was written to any flash control register and generates a PUC when set. KEYV must be reset with software. KEYV is reset with POR.<br>0b = FCTLx password was written correctly<br>1b = FCTLx password was written incorrectly                                                |
| Bit | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                               |
| 0   | BUSY    | R    | 0h    | Busy. This bit indicates the status of the flash timing generator.<br>0b = Not busy<br>1b = Busy                                                                                                                                                                                                                                                          |

## Flash Control Registers

**Figure 7-16. FCTL4 Register**

| 15       | 14   | 13   | 12   | 11       | 10   | 9    | 8    |
|----------|------|------|------|----------|------|------|------|
| FWKEYx   |      |      |      |          |      |      |      |
| rw-1     | rw-0 | rw-0 | rw-1 | rw-0     | rw-1 | rw-1 | rw-0 |
| 7        | 6    | 5    | 4    | 3        | 2    | 1    | 0    |
| Reserved |      | MRG1 | MRG0 | Reserved |      |      |      |
| r-0      | r-0  | rw-0 | rw-0 | r-0      | r-0  | r-0  | r-0  |

**Table 7-9. FCTL4 Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                  |
|------|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | FWKEYx   | R/W  | 96h   | FCTLx password. Always reads as 096h. Must be written as 0A5h. Writing any other value generates a PUC.                                                                                                                                                                                                      |
| 7-6  | Reserved | R    | 0h    | Reserved. Always read as 0.                                                                                                                                                                                                                                                                                  |
| 5    | MRG1     | R/W  | 0h    | Marginal read 1 mode. This bit enables the marginal 1 read mode. The marginal read 1 bit is cleared if the CPU starts execution from the flash memory. If both MRG1 and MRG0 are set, MRG1 is active and MRG0 is ignored.<br>0b = Marginal 1 read mode is disabled.<br>1b = Marginal 1 read mode is enabled. |
| 4    | MRG0     | R/W  | 0h    | Marginal read 0 mode. This bit enables the marginal 0 read mode. The marginal mode 0 is cleared if the CPU starts execution from the flash memory. If both MRG1 and MRG0 are set, MRG1 is active and MRG0 is ignored.<br>0b = Marginal 0 read mode is disabled.<br>1b = Marginal 0 read mode is enabled.     |
| 3-0  | Reserved | R    | 0h    | Reserved. Always read as 0.                                                                                                                                                                                                                                                                                  |

Flash Memory Control 4 Register. This register is not available in all devices. See the device-specific data sheet for details.

## Flash related in MSP430.h

---

```

/*****
* Flash Memory
*****/
#define __MSP430_HAS_FLASH2__          /* Definition to show that Module is available
*/

SFR_16BIT(FCTL1);                      /* FLASH Control 1 */
SFR_16BIT(FCTL2);                      /* FLASH Control 2 */
SFR_16BIT(FCTL3);                      /* FLASH Control 3 */

#define FRKEY                          (0x9600) /* Flash key returned by read */
#define FWKEY                          (0xA500) /* Flash key for write */
#define FXKEY                          (0x3300) /* for use with XOR instruction */
#define ERASE                          (0x0002) /* Enable bit for Flash segment erase */
#define MERAS                          (0x0004) /* Enable bit for Flash mass erase */
#define WRT                            (0x0040) /* Enable bit for Flash write */
#define BLKWRT                         (0x0080) /* Enable bit for Flash segment write */
#define SEGWRT                         (0x0080) /* old definition */ /* Enable bit for Flash
segment write */

```



## Flash related in MSP430.h

---

```
#define FSSEL_0          (0x0000)    /* Flash clock select: 0 - ACLK */
#define FSSEL_1          (0x0040)    /* Flash clock select: 1 - MCLK */
#define FSSEL_2          (0x0080)    /* Flash clock select: 2 - SMCLK */
#define FSSEL_3          (0x00C0)    /* Flash clock select: 3 - SMCLK */

#define BUSY              (0x0001)    /* Flash busy: 1 */
#define KEYV              (0x0002)    /* Flash Key violation flag */
#define ACCVIFG           (0x0004)    /* Flash Access violation flag */
#define WAIT              (0x0008)    /* Wait flag for segment write */
#define LOCK              (0x0010)    /* Lock bit: 1 - Flash is locked (read only) */
#define EMEX              (0x0020)    /* Flash Emergency Exit */
#define LOCKA             (0x0040)    /* Segment A Lock bit: read = 1 - Segment is
locked (read only) */
#define FAIL              (0x0080)    /* Last Program or Erase failed */
```



## Port related in MSP430.h

---

```
#define BIT0      (0x0001)
#define BIT1      (0x0002)
#define BIT2      (0x0004)
#define BIT3      (0x0008)
#define BIT4      (0x0010)
#define BIT5      (0x0020)
#define BIT6      (0x0040)
#define BIT7      (0x0080)
#define BIT8      (0x0100)
#define BIT9      (0x0200)
#define BITA      (0x0400)
#define BITB      (0x0800)
#define BITC      (0x1000)
#define BITD      (0x2000)
#define BITE      (0x4000)
#define BITF      (0x8000)
```





## THANK YOU

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**Rex Joseph**

Department of Electrical and Electronics Engineering

**[rexjoseph@pes.edu](mailto:rexjoseph@pes.edu)**

+91 80 6666 3333 Extn 515